Farming problems and the carbon cycle

9B Plant growth · Lesson 6 of 7

Date _____ Class _____

Today I can

- ☐ I can explain how fertilisers, pesticides and clearing land can harm wild organisms, including eutrophication and bioaccumulation.
- ☐ I can use the carbon cycle to explain how farming and burning fuels change the carbon dioxide in the air, and suggest ways to farm more sustainably.

What happens today

1. Do Now: the dead fish
2. Learn: Eutrophication · Pesticide problems · and more
3. Core tasks
4. Challenge, then the exit ticket

Student Book pages 32 to 33 · Workbook pages 23 to 24

What I learned today (one or two sentences)

Key Vocabulary

Term	Definition
habitat	the place where an organism lives
eutrophication	a very rich supply of nutrients in water, which leads to fast plant growth and then death of fish
algae	simple plant-like organisms that photosynthesise in water
decomposer	a microorganism that breaks down dead material, using oxygen
persistent	does not break down in the environment
bioaccumulation	the build-up of a substance in organisms along a food chain
food web	a diagram of linked food chains showing what eats what
biodiversity	the number of different species in an area
carbon cycle	the processes that move carbon between the air, living things and the Earth
combustion	burning, which releases carbon dioxide
global warming	the rise in the Earth's average temperature
sustainable development	making the things we need without destroying habitats for the future

Quick Review

the dead fish

1 What do fish need from the water for respiration?

2 Name one farm chemical that rain can wash into a river or lake.

3 Suggest how these fish died.

Core 1 of 4

Everyone does these. Answer on the card or under it.

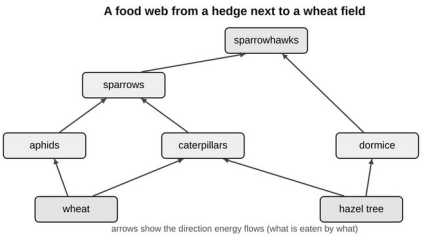
1 Number the statements 1 to 6 to show how using fertiliser can lead to the death of fish. The first and last have been done for you.

Order	Statement
1	Nutrients from the fertiliser dissolve in soil water.
	Decomposers break down the dead plants, using oxygen from the water.
	Extra nutrients in the water cause rapid growth of algae and water plants.
	Soil water with extra nutrients leaks into a nearby stream or river.
	Algae and water plants at the surface block the light, so deeper plants die.
6	Fish die because there is too little oxygen in the water.

Core 2 of 4

Everyone does these. Answer on the card or under it.

2 The diagram shows a food web from a hedge next to a wheat field.



a) Name one producer in this food web.

Core 3 of 4

Everyone does these. Answer on the card or under it.

2 *continued*

b) Name two animals that eat caterpillars.

c) The farmer sprays the wheat with a general insecticide, which kills all insects. Tick one box in each row to show what happens to each population.

Population	Increases	Decreases	No change
aphids			
caterpillars			
sparrows			
sparrowhawks			

Core 4 of 4

Everyone does these. Answer on the card or under it.

3 Match each process in the carbon cycle to what it does. Draw a line.

photosynthesis	releases carbon dioxide when fuels or wood burn
respiration	passes carbon compounds from plants to animals
combustion	removes carbon dioxide from the air
decomposers feeding	releases carbon dioxide from dead plants and animals
animals feeding	releases carbon dioxide from living cells, day and night

Challenge 1 of 2

When Core is finished. Full sentences.

C1 In 19th century Ireland, most people survived by growing and eating one variety of potato, called the lumper. Between 1845 and 1852 more than a million people starved. Suggest why, using the words variety, disease and biodiversity.

Challenge 2 of 2

When Core is finished. Full sentences.

C2 A persistent insecticide was sprayed on crops next to a river for many years. Years later, fish-eating birds at the top of the river food chain were found to contain large amounts of the insecticide.

a) Explain how the insecticide reached the birds and why they contained so much of it.

b) In 2014 the EU banned some neonicotinoid insecticides on rapeseed, which flowers, but not on wheat.

Suggest why.

Exit Ticket

Hand this card in at the end of the lesson.

Name _____

1 What does eutrophication mean?

2 Name two processes that release carbon dioxide into the air.

3 Why is it risky to plant one variety of crop over a very large area?

Today I can: Goal 1 ☐ Goal 2 ☐